

SIMATS ENGINEERING

CO2

ASSESSMENT TOOL 3

COURSE NAME: CSA07

NAME : Lavanya M

COURSE CODE: Computer Networks

REG NO : 192521072

Experiment: IPv6 Configuration and Debugging in Cisco Packet Tracer

Objective :

To configure IPv6 addresses on two devices, verify subnet membership, identify addressing issues, optimize the IPv6 configuration, and test communication using Cisco Packet Tracer.

Step 1: Network Topology

Description:

- Added two PCs (Device A and Device B) and one 2960 switch.
- Connected both PCs to the switch using Copper Straight-Through cables.
- Verified that all links turned green, indicating successful physical connectivity.

Step 2: IPv6 Configuration

Description:

- Configured static IPv6 addresses on both PCs.
- Assigned a prefix length of 64 to each device.
- Left the default gateway blank since both devices are connected within the same local network.
- Configured Addresses:
 - Device A: 2001:db8:0:0::1/64
 - Device B: 2001:db8:0:0::2/64

Step 3: Connectivity Test

Description:

- Opened the Command Prompt on Device A.
- Used the ping command to test communication with Device B.

- Received four successful replies, confirming that both devices can communicate over IPv6.

Command Used:

ping 2001:db8:0:0::2

Observations

The IPv6 addresses were configured successfully.

Both devices belong to the same /64 subnet.

Physical connectivity was verified through the switch.

The ping test was successful with four replies received.

Result

The IPv6 network was successfully configured in Cisco Packet Tracer. Both devices communicated successfully after the correct IPv6 configuration was applied, verifying proper IPv6 addressing and connectivity.

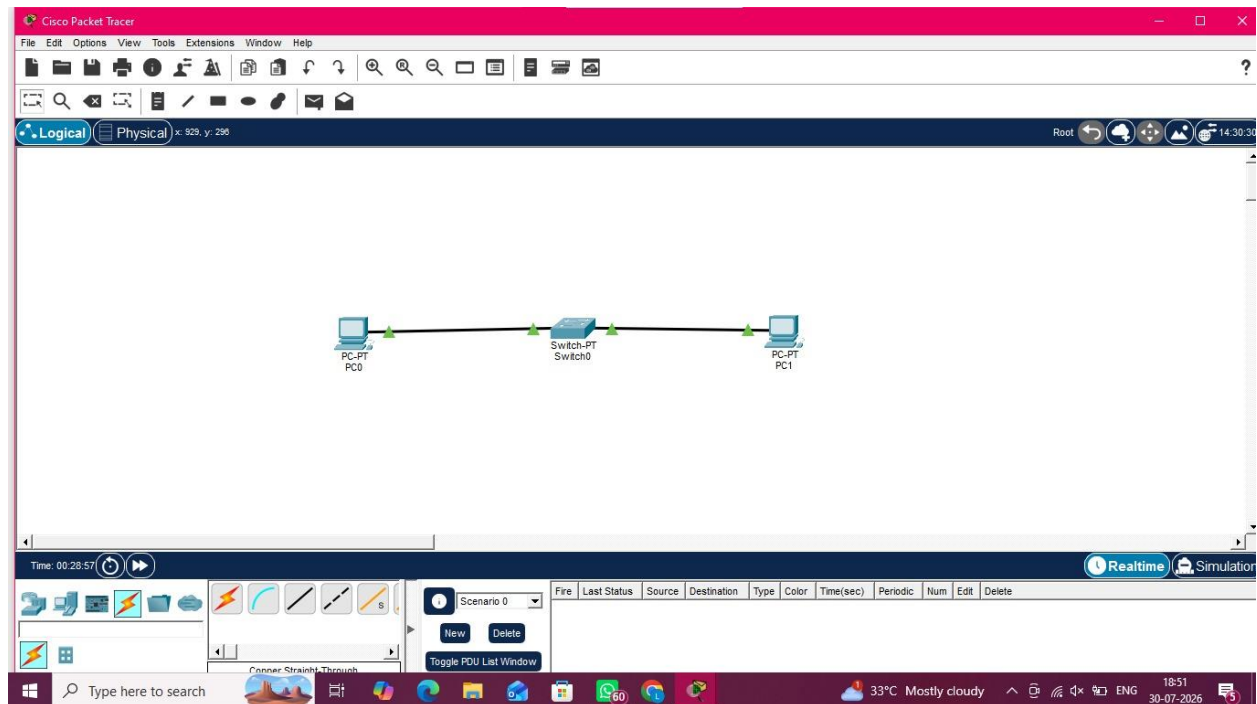


Figure 1: Network topology with two PCs connected through a switch

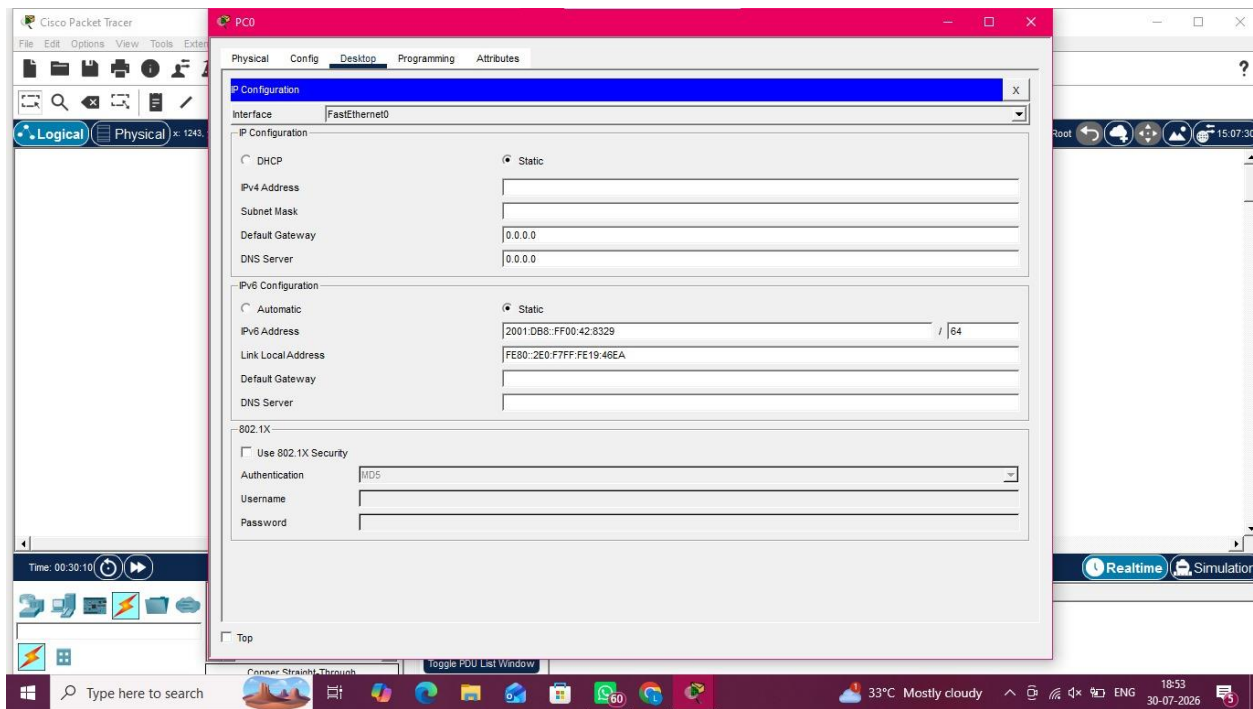


Figure 2: IPv6 configuration of Device A.

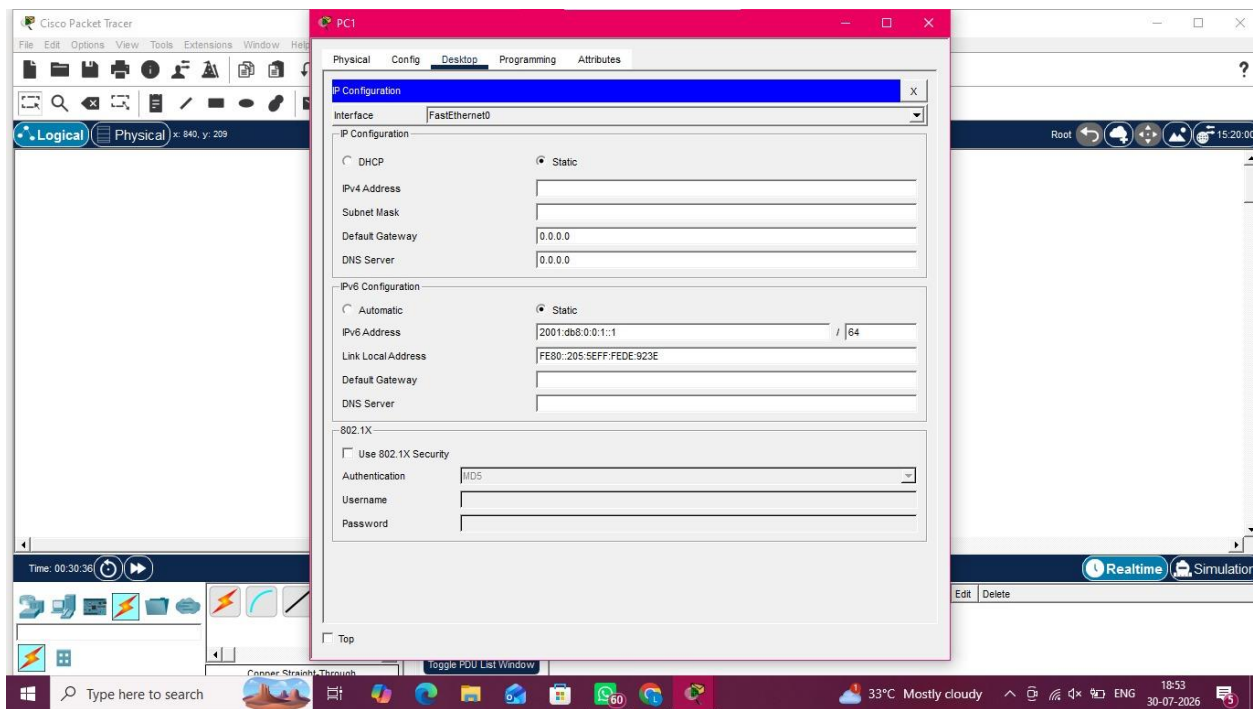


Figure 3: IPv6 configuration of Device B.

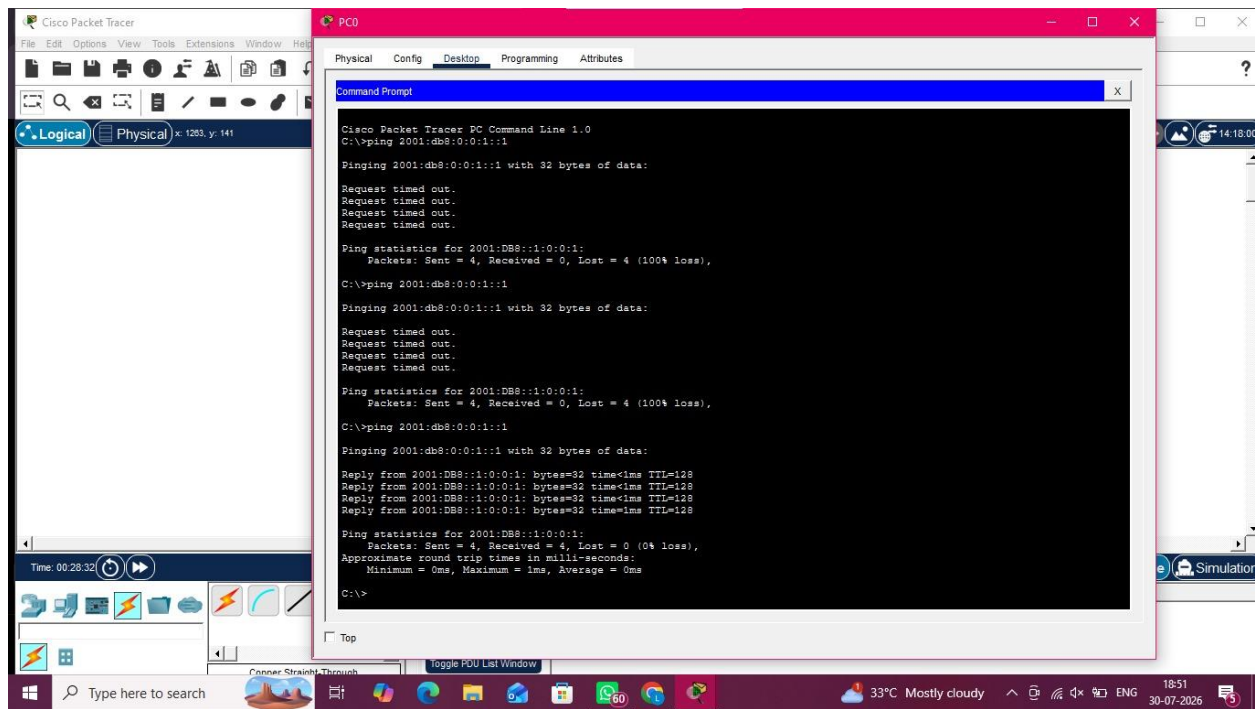


Figure 4: Successful IPv6 communication verified using the ping command.